

The econ-viz Package: Publication-Quality Microeconomics Diagrams

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1 Introduction

The `econ-viz` package is a Python toolkit for producing publication-quality microeconomics diagrams: indifference curves, budget constraints, consumer equilibria, demand curves and exchange economies. It draws in the conventions of textbook figures—first-quadrant axes with arrow tips, L^AT_EX-style labels and no numeric ticks—and exports to PNG, PDF, SVG, TikZ or animated GIF.

This manual describes v1.12.0; installation is covered in Section 2 and a first example in Section 3.

1.1 Scope

`econ-viz` is organised in five parts: utility models, equilibrium solving, utility levels, drawing layers, and output formats. Models and solutions remain usable on their own, and the same model can be passed to a canvas, multi-panel figure, demand diagram, or analysis API.

1.2 Reading guide

The manual is organised in four parts (Table 1):

Table 1
Chapter guide

Topic	Subtopic	Description	Section
Drawing & export	Canvas & layers	Axes, curves, equilibria	Section 4
	Figures & demand	Panels & demand	Section 5
	Themes & config	Colours, styles, config	Section 9
	Output formats	Image files and TikZ	Section 10
Models & analysis	Built-in models	Utility functions	Section 6
	Custom & many-good	Custom, N goods	Section 7
	Comparative statics	Derivatives & Slutsky	Section 8
Animation & interaction	GIF animation	Sweeps as GIFs	Section 12
	Jupyter widgets	Sliders in notebooks	Section 13
Other inputs	Command line	Plots from the shell	Section 11
	L ^A T _E X parser	Models from formulas	Section 14

On first use, read Section 3 and Section 4 in order. To look up a particular API, go straight to its section or to the command index.

¹The Python website provides installers for every operating system: <https://www.python.org/downloads/>.

2 Installation

2.1 Requirements

econ-viz requires Python 3.10 or later¹. This chapter manages packages and projects with uv² and gives the matching pip command where useful.

Installing the package also installs NumPy, SciPy, matplotlib and SymPy, for array computation, numerical solving, plotting and symbolic algebra³.

2.2 Installing uv

The commands in this manual use uv. Existing projects can keep using pip, pipx or Poetry the package API does not depend on the tool that installed it.

Run the installer for your operating system.

macOS, Linux

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Windows

```
powershell -ExecutionPolicy Bypass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

2.3 Installing econ-viz

Create a project and add econ-viz as a dependency:

```
uv init my-diagrams
cd my-diagrams
uv add econ-viz
```

Run scripts inside the project environment with uv run:

```
uv run python main.py
```

Save the basic example of Section 3 as main.py in the project directory, then run the command above. uv add records the dependency and uv run uses the project's Python environment; install and run in the same environment.

To pin the version this manual describes, give it when adding the dependency:

```
uv add "econ-viz==1.12.0"
```

In an existing Python virtual environment, install it with pip:

```
python -m pip install -U econ-viz
```

The package is named econ-viz; its Python import name is econ_viz:

```
from econ_viz import Canvas, solve
```

An import statement cannot contain a hyphen. On ModuleNotFoundError, check that the interpreter running the script is the one the package was installed into.

2.4 Optional dependencies

Animation and Jupyter widgets need optional dependencies. Install the group you need:

animation	GIF export with Animator (Section 12); pulls in Pillow.
interactive	Notebook widgets with WidgetViewer (Section 13); pulls in ipywidgets and IPython.
all	Every optional dependency.

²uv is a fast Python package and project manager by Astral that can also manage Python versions; see its documentation for installation and usage: <https://docs.astral.sh/uv/>.

³Ordinary Python plotting needs no L^AT_EX installation; a L^AT_EX distribution is only required to compile exported TikZ code.

```
uv add "econ-viz[animation]" # GIF export (Pillow)
uv add "econ-viz[interactive]" # notebook widgets
uv add "econ-viz[all]" # all extras
```

Installing an extra does not run an animation or open a notebook by itself; call the API as shown in the relevant chapter.

2.5 Installing the command-line tool

To use only the command-line interface, install `econ-viz` as a standalone tool (Section 11):

```
uv tool install econ-viz
```

This installs the tool in an environment of its own. To import the package in your own Python code, still run `uv add econ-viz` in that project; calling `uv run econ-viz` inside the project keeps the CLI and your code on the same version.

2.6 Development setup

```
git clone https://github.com/EconViz/econ-viz.git
cd econ-viz
uv sync --all-extras
```

`uv sync --all-extras` installs the development dependencies and every optional dependency. Then run the tests:

```
uv run pytest
```

2.7 Verifying the installation

```
uv run econ-viz --version # econ-viz 1.12.0
uv run econ-viz help
```

The version in the comment is only an example; the output reflects the installed version. To check that Python can import the drawing and solving API:

```
uv run python -c "from econ_viz import Canvas, solve; print('OK')"
```

On a server or any environment without a display, write files with `save()` or the CLI's `--output`. `show()` needs an interactive plotting backend, so a window that fails to open does not mean the installation failed.

3 Quick start

3.1 A minimal example

This chapter uses the complete Cobb-Douglas consumer problem:

$$\begin{aligned} \max_{x,y} u(x,y) &= x^{\frac{1}{2}}y^{\frac{1}{2}} \\ \text{s.t. } 2x + 3y &= 30 \end{aligned}$$

The script solves the optimum and draws the indifference map, budget set, and equilibrium shown in Figure 1.

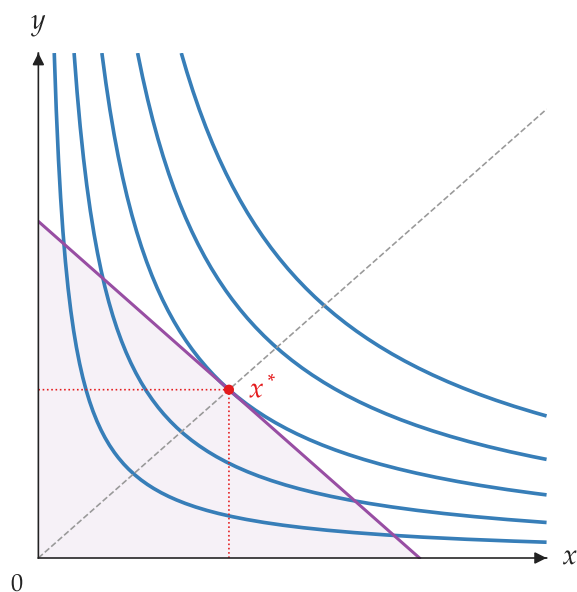


Figure 1
The consumer-equilibrium diagram.

Example 1

```
from econ_viz import Canvas, levels, solve
from econ_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)
lvls = levels.around(eq.utility, n=5)

cvs = Canvas(
    x_max=20, y_max=15,
    x_label="x", y_label="y",
    title=r"Cobb-Douglas  $x^{0.5} y^{0.5}$ "
)
cvs.add_utility(model, levels=lvls)
cvs.add_budget(2.0, 3.0, 30.0, fill=True)
cvs.add_equilibrium(eq, show_ray=True)
cvs.save("cobb_douglas.png")
```

Running `main.py` writes `cobb_douglas.png` to the current working directory. The optimum is $x^* = 7.5$, $y^* = 5$; spending is $2 \times 7.5 + 3 \times 5 = 30$, exactly the income. The budget line meets the axes at $\frac{I}{p_x} = 15$ and $\frac{I}{p_y} = 10$.

3.2 Step by step

Every diagram follows the same steps: build a model, solve for the equilibrium, pick utility levels, create a canvas, add layers and export.

```
econ_viz.models from econ_viz.models import CobbDouglas
model = CobbDouglas(alpha=0.5, beta=0.5)
```

Pick a utility function. Section 6 lists every built-in model; any Python function can also be wrapped as a model (Section 7).

```
solve solve(<model>, px=<float>, py=<float>, income=<float>)
      Solve the consumer's problem and return an immutable Equilibrium.
x      Optimal quantity  $x^*$ .
float
y      Optimal quantity  $y^*$ .
float
utility Utility at the optimum.
float
```

- `bundle_type` Solution type: "interior" for an interior solution, "boundary" when the numerical solution sits at a quantity lower bound (e.g. Stone-Geary subsistence), "kink" for a Leontief kink and "corner" for a perfect-substitutes corner.
- ```

from econ_viz import solve
eq = solve(model, px=2.0, py=3.0, income=30.0)
print(eq.x, eq.y, round(eq.utility, 3))

7.5 5.0 6.124

```
- `levels.around` `levels.around(<utility>, n=<int>)`
- Generate  $n$  utility levels centred on a given utility. Figure 1 uses  $n=5$ , so it draws five indifference curves.
- `levels.around()` keeps the given utility among the levels, so the equilibrium lies on one of the curves. Passing `levels=5` instead lets the canvas pick levels from the sampled utility range, which need not include the equilibrium utility. To compare the same preferences across several figures, compute `lvls` once and pass it to each figure.
- `Canvas` `Canvas(x_max=<float>, y_max=<float>, ...)`
- Create the canvas and set the axis ranges; the other options are described in Section 4.
- The axis ranges only control what is shown; they play no part in `solve()`. If the equilibrium is missing from the figure, check whether `eq.x` or `eq.y` lies outside the range and raise the axis limits.
- `Canvas.add_*` Add indifference curves, a budget line or the equilibrium to the same canvas. The calls can be chained (Section 4.4). `add_equilibrium()` uses an existing solution and does not solve again; after changing prices, income or the model, call `solve()` again and build a new figure.
- `Canvas.save` `save()` writes the file in the format given by its extension (Section 10); `show()` opens a
- `Canvas.show` `matplotlib` window. Output paths are relative to the working directory of the script; create any subdirectory beforehand.

### Checking inputs and failed solves

Prices and income must be positive; otherwise `solve()` raises `InvalidParameterError`<sup>4</sup>.

When the solver does not converge, `solve()` raises `OptimizationError` rather than returning an invalid equilibrium.

```

from econ_viz import InvalidParameterError, OptimizationError, solve

try:
 eq = solve(model, px=2.0, py=3.0, income=30.0)
except (InvalidParameterError, OptimizationError) as error:
 print(error)
else:
 print(round(eq.x, 3), round(eq.y, 3), eq.bundle_type)

```

Numerical solutions carry small errors: a result may be close to, but not exactly, 7.5. Compare with a tolerance when checking results, and round with `round()` only for display, so early rounding does not distort later calculations.

## 4 Canvas

Canvas builds on a `matplotlib` Figure and Axes and provides layers for indifference curves, budget lines, equilibria and more.

---

<sup>4</sup>Smooth models are solved with sequential least squares programming (SLSQP).

## 4.1 Constructor

```
Canvas from econ_viz import ArrowStyle, Canvas, Stroke, themes

cvs = Canvas(
 x_max=20,
 y_max=15,
 x_label="x",
 y_label="y",
 title=r"Cobb-Douglas $x^{0.5} y^{0.5}$ ",
 dpi=300,
 x_label_pos="right", # "top", "right" or "bottom"
 y_label_pos="top", # "left", "top" or "right"
 font="DejaVu Sans",
 math_font="stix",
 axis_stroke=Stroke(width=1.0, arrow=ArrowStyle.TRIANGLE),
 theme=themes.default,
)
```

*Updated: v1.7.0* Create a canvas and set its axis ranges, axis labels, fonts, line styles and theme. Visual settings not passed explicitly come from the active Config and Theme.

|                                                          |                                                                                                                  |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <code>x_max</code><br><i>float · default 10</i>          | Upper bound of the horizontal axis.                                                                              |
| <code>y_max</code><br><i>float · default 10</i>          | Upper bound of the vertical axis.                                                                                |
| <code>x_label</code><br><i>str · default "x"</i>         | Label at the tip of the horizontal axis.                                                                         |
| <code>y_label</code><br><i>str · default "y"</i>         | Label at the tip of the vertical axis.                                                                           |
| <code>title</code><br><i>str   None · default None</i>   | Figure title.                                                                                                    |
| <code>dpi</code><br><i>int · default 300</i>             | Raster export resolution, clamped to 1–1200.                                                                     |
| <code>x_label_pos</code><br><i>str · default "right"</i> | Position of the horizontal label relative to the arrow; a <code>LabelPosition</code> is also accepted (Table 2). |

**Table 2**  
x\_label\_pos values

| Value    | Position           |
|----------|--------------------|
| "top"    | Above the arrow    |
| "right"  | Right of the arrow |
| "bottom" | Below the arrow    |

`y_label_pos`  
*str · default "top"* Position of the vertical label relative to the arrow (Table 3).

**Table 3**  
y\_label\_pos values

| Value   | Position           |
|---------|--------------------|
| "left"  | Left of the arrow  |
| "top"   | Above the arrow    |
| "right" | Right of the arrow |

`font`  
*str | list · default None* Font for all text, or a list of candidate fonts. It applies to this canvas only and leaves matplotlib's global configuration alone.

`math_font`  
*str · default None* matplotlib math font set: dejavusans, dejavuserif, cm, stix, stixsans and so on.

|                                             |                                                                |
|---------------------------------------------|----------------------------------------------------------------|
| <code>axis_stroke</code>                    | Width, style, colour and arrowhead of both axes (Section 4.3). |
| <code>Stroke · default theme</code>         |                                                                |
| <code>x_axis_stroke</code>                  | Override for the horizontal axis.                              |
| <code>Stroke · default None</code>          |                                                                |
| <code>y_axis_stroke</code>                  | Override for the vertical axis.                                |
| <code>Stroke · default None</code>          |                                                                |
| <code>theme</code>                          | Colour and style theme (Section 9).                            |
| <code>Theme · default themes.default</code> |                                                                |

## 4.2 Drawing methods

The `add_*` methods add a layer to the canvas and return the same Canvas, so calls can be chained (Section 4.4).

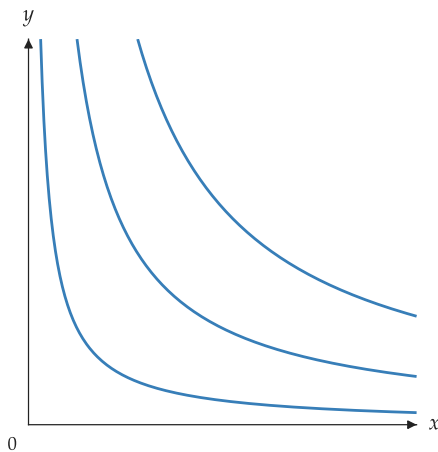
```
add_utility cvs.add_utility(
 func,
 levels=3, # count or list of levels
 color=None, # default: theme.ic_color
 linewidth=None, # default: theme.ic_linewidth
 show_rays=False,
 show_kinks=False,
 kink_radius=1.0,
 show_bliss=True, # star at the bliss point (Satiation)
 show_ic_labels=False,
 stroke=None,
 ray_stroke=None,
 ic_label=None,
 highlight_level=None,
 secondary_stroke=None,
 label_style="numeric",
)
```

Draw indifference curves for a utility model.

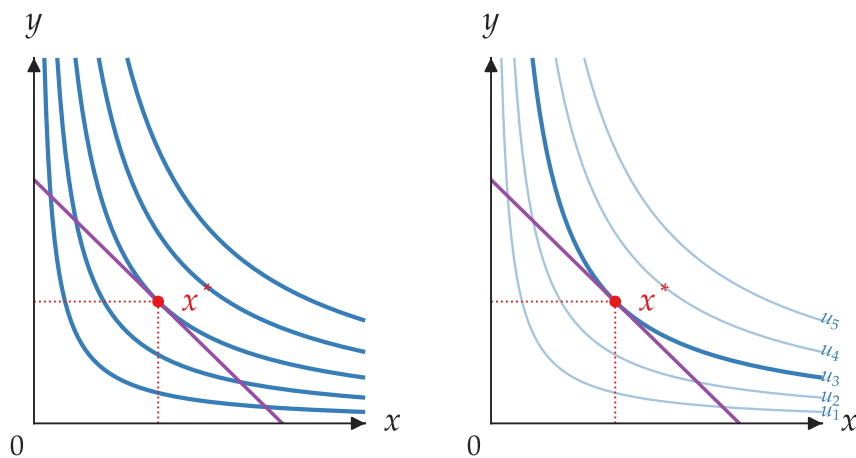
|                                           |                                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <code>levels</code>                       | A number of curves, or a list of utility values such as <code>levels.around(eq.utility, n=5)</code> to place them around the optimum. |
| <code>int   list · default 3</code>       |                                                                                                                                       |
| <code>highlight_level</code>              | Focal utility level. The closest curve is drawn in the main style.                                                                    |
| <code>float   None · default None</code>  |                                                                                                                                       |
| <code>secondary_stroke</code>             | Style of the other curves; defaults to <code>theme.secondary_ic_stroke</code> .                                                       |
| <code>Stroke   None · default None</code> |                                                                                                                                       |
| <code>show_ic_labels</code>               | Label the curves. Labels follow the curve's tangent and stay inside the plotting area (Figure 3).                                     |
| <code>bool · default False</code>         |                                                                                                                                       |
| <code>label_style</code>                  | "numeric" shows utility values; "ordinal" numbers the curves $u_1, u_2, \dots$ in increasing utility.                                 |
| <code>str · default "numeric"</code>      |                                                                                                                                       |

```
lvls = levels.around(eq.utility, n=5)
cvs.add_utility(
 model,
 levels=lvls,
 highlight_level=eq.utility,
 show_ic_labels=True,
 label_style="ordinal",
)
```

```
add_budget cvs.add_budget(
 px, py, income,
 color=None,
 linewidth=None,
 linestyle="-",
 label=None, # legend label (LaTeX)
 fill=False, # shade feasible set
 fill_alpha=None, # default: theme.budget_fill_alpha
)
```



**Figure 2**  
Three indifference curves.



**Figure 3**  
Focal and secondary curves.

```
stroke=None,
)
```

Draw the budget line  $p_x x + p_y y = I$ .

`fill` `True` shades the feasible set with the theme's fill; pass a `Fill` to set its colour and opacity.  
*bool* | *Fill* · default `False`

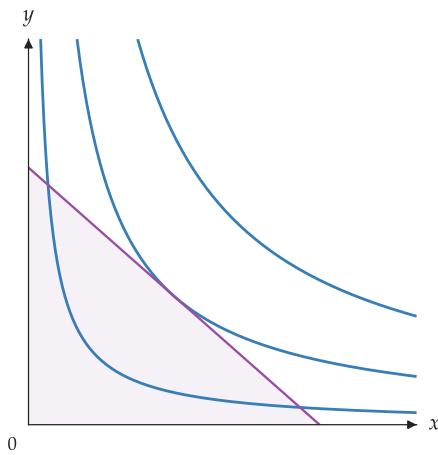
`stroke` Colour, width, style and opacity of the line in one object; takes precedence over `color`, `linewidth` and `linestyle` (Section 4.3).  
*Stroke* | *None* · default `None`

`label` Legend text, shown once `show_legend()` is called.  
*str* | *None* · default `None`

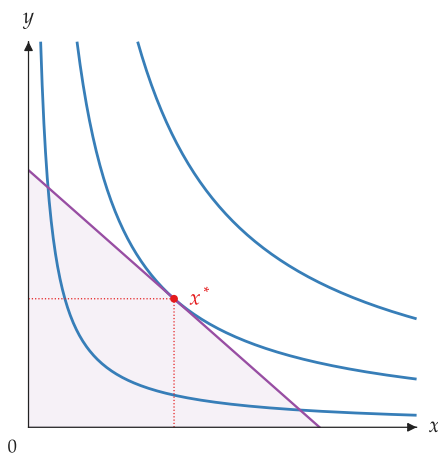
```
add_equilibrium cvs.add_equilibrium(
 eq, # result of solve()
 color=None,
 markersize=None,
 label="x^*",
 drop_dashes=True, # dashed lines to axes
 show_ray=False, # expansion path
 drop_stroke=None,
 ray_stroke=None,
 marker=None,
)
```

Mark the optimal bundle returned by `solve()`.

`drop_dashes` Drop lines to both axes.  
*bool* · default `True`



**Figure 4**  
Budget line and shading.



**Figure 5**  
Equilibrium and drop lines.

`show_ray` Draw the expansion path through the origin and the optimum.  
*bool · default False*

`marker` Style of the equilibrium marker (Section 4.3).  
*Marker | None · default None*

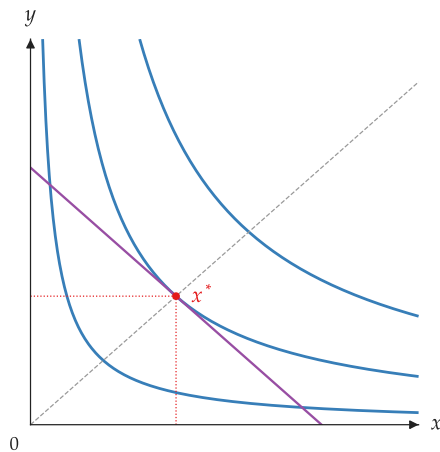
`add_ray` `cvs.add_ray(`  
    `slope,` #  $dy/dx$   
    `color=None,`  
    `linewidth=None,`  
    `stroke=None,`  
)

Draw a ray from the origin with slope  $slope$  ( $dy/dx$ ), for an expansion path or a fixed consumption ratio.

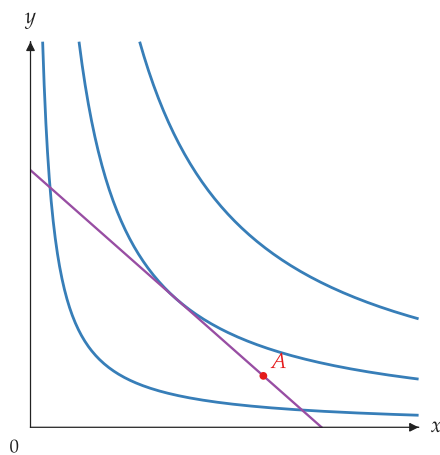
`add_point` `cvs.add_point(`  
    `x, y,`  
    `label=None,`  
    `color=None,`  
    `markersize=None,`  
    `offset=None,` # label offset (pt)  
    `marker=None,`  
)

Mark any point, such as a bundle to compare with the optimum.

`label` Marker text, as a string or a Label.  
*str | Label | None · default None*



**Figure 6**  
Expansion-path ray.

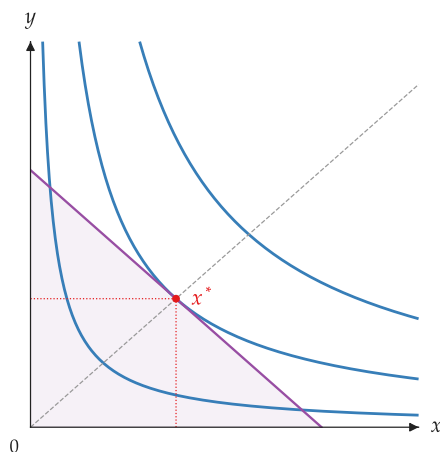


**Figure 7**  
A labelled point A.

`offset` Offset of the text from the point, in points.  
*tuple* | *None* · default *None*  
`marker` Style of the point marker.  
*Marker* | *None* · default *None*

```
show cvs.show() # interactive window
save cvs.save("figure.png") # .png / .pdf / .svg / .tex
```

`show()` opens an interactive window. `save()` writes the file in the format given by its extension (Section 10) and releases the `matplotlib` figure, so call it last.



**Figure 8**  
The complete canvas.

### 4.3 Styles

LineStyle defines line patterns, ArrowStyle arrowheads, and Stroke bundles width, pattern, colour and arrowhead for axes, budget lines, paths and drop lines.

#### Line styles

LineStyle LineStyle.SOLID, DASHED, DOTTED, DASHDOT

*New: v1.7.0* Solid, dashed, dotted and dash-dot lines (Figure 9). Axes accept the enum or the matching string; other lines use Stroke(style=...).

```
import matplotlib.pyplot as plt

from econ_viz import Canvas, LineStyle

styles = [
 LineStyle.SOLID,
 LineStyle.DASHED,
 LineStyle.DOTTED,
 LineStyle.DASHDOT,
]

fig, axes = plt.subplots(1, 4, figsize=(8.5, 2.2))
for ax, style in zip(axes, styles):
 Canvas(
 title=style.value,
 x_line_style=style,
 y_line_style=style,
 fig=fig,
 ax=ax,
)

Axis labels sit outside the arrowheads; give four panels more room.
fig.subplots_adjust(wspace=0.62)
fig.savefig("line_styles.png", dpi=160, transparent=True)
```

#### Arrow styles

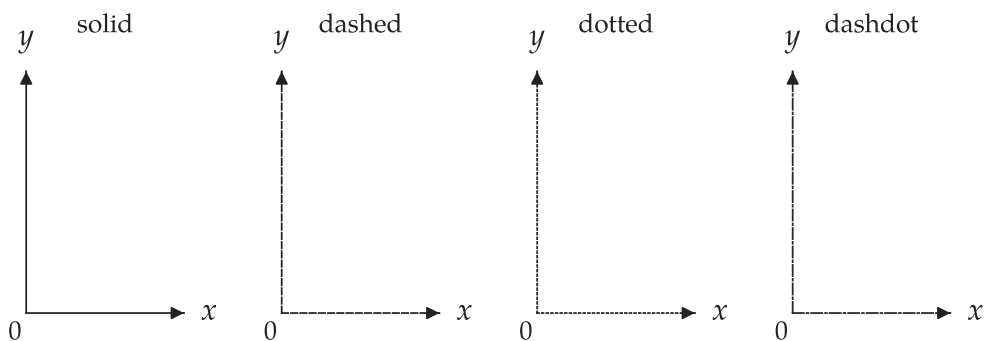
ArrowStyle ArrowStyle.SIMPLE, TRIANGLE, FANCY, WEDGE

*New: v1.7.0* Four arrowhead styles (Figure 10). Axes use x\_arrow\_style and y\_arrow\_style; other lines use Stroke(arrow=...).

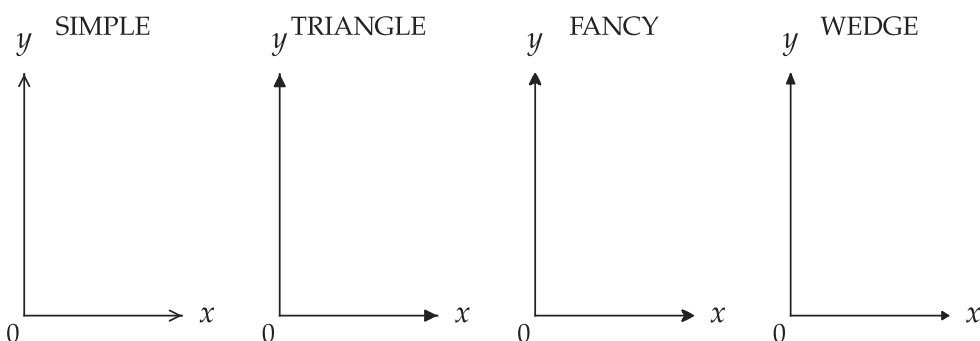
#### Stroke

Stroke Stroke(width=<float>, style=<LineStyle>, color=<str>, arrow=<ArrowStyle>, opacity=<float>)

*New: v1.7.0* Width, pattern, colour, arrowhead and opacity of one line. Fields left unset inherit from the active theme. *Updated: v1.9.0* opacity ranges from 0 to 1.



**Figure 9**  
The four line styles.



**Figure 10**  
The four arrow styles.

### Axis

**Axis** `Axis(label=None, label_position=None, stroke=None)`

*New: v1.8.0* Label, label position and line of one axis in one object. `Canvas`, `Figure`, `DemandDiagram` and `EdgeworthBox` all accept `x_axis` and `y_axis`. When shorthand arguments are passed as well, fields set in the `Axis` win.

```
from econ_viz import Axis, Canvas, Label, Stroke

cvs = Canvas(
 x_axis=Axis(
 label=Label(text="x_1", fontsize=16),
 label_position="bottom",
 stroke=Stroke(width=1.2),
),
 y_axis=Axis(label="x_2"),
)
```

### Marker and Label

**Marker** `Marker(color=None, size=None, shape=None, opacity=None)`

*New: v1.8.0* Colour, size, shape and opacity of a point marker. `shape` takes a matplotlib marker string such as "o", "s", "^", "D" or "\*".

**Label** `Label(text=None, position=None, offset=None, color=None, fontsize=None, visible=None, opacity=None)`

*New: v1.8.0* Text, position, offset, colour, size, visibility and opacity of a label. The position is a side or a corner; `Label(visible=False)` hides the text.

*Updated: v1.9.0*

```
from econ_viz import Label, Marker

cvs.add_equilibrium(
 eq,
 marker=Marker(shape="s", size=8),
 label=Label(position="bottom-left", offset=8),
)
cvs.add_point(
 12, 2,
 marker=Marker(color="black", shape="D"),
 label=Label(text="A", position="left", fontsize=14),
)
```

### Fill

**Fill** `Fill(color=None, opacity=None)`

*New: v1.8.0* Colour and opacity of a shaded area. `alpha` is a compatible shorthand for opacity; one call must not pass different values for both.

```
from econ_viz import Fill

cvs.add_budget(
```

```

 2, 3, 30,
 fill=Fill(color="lightgrey", opacity=0.4),
)

```

## 4.4 Method chaining

The `add_*` methods can be chained, ending with `save()`:

```

Canvas(x_max=20, y_max=15) \
 .add_utility(model, levels=lvls) \
 .add_budget(2.0, 3.0, 30.0, fill=True) \
 .add_equilibrium(eq, show_ray=True) \
 .save("figure.png")

```

# 5 Figures and demand diagrams

On top of `Canvas`, `econ-viz` builds these larger diagrams:

- `Figure` for multi-panel layouts;
- `PricePath` and `IncomePath` for budget and equilibrium sweeps;
- `DemandDiagram` for linked goods-space and Marshallian-demand views;
- `decompose_price_effect` for the Hicks and Slutsky decompositions;
- `EdgeworthBox` for two-consumer exchange diagrams.

## 5.1 Multi-panel figures

**Figure** `Figure(<layout>, x_max=<float>, y_max=<float>, ..., shared_x=False, shared_y=False)`  
*New: v1.2.0* Compose several canvases into one figure, for before-and-after comparisons, decompositions or classroom slides. It takes the same styling arguments as `Canvas`. `fig[idx]` returns the panel `Canvas`, so the drawing API is unchanged once the layout exists.

### Example 2

```

from econ_viz import Figure, Layout, Legend, levels, solve
from econ_viz.models import CobbDouglas

fig = Figure(
 Layout.SIDE_BY_SIDE,
 x_max=20,
 y_max=15,
 x_label="x",
 y_label="y",
 title="Before / After Price Change",
 shared_y=True,
)

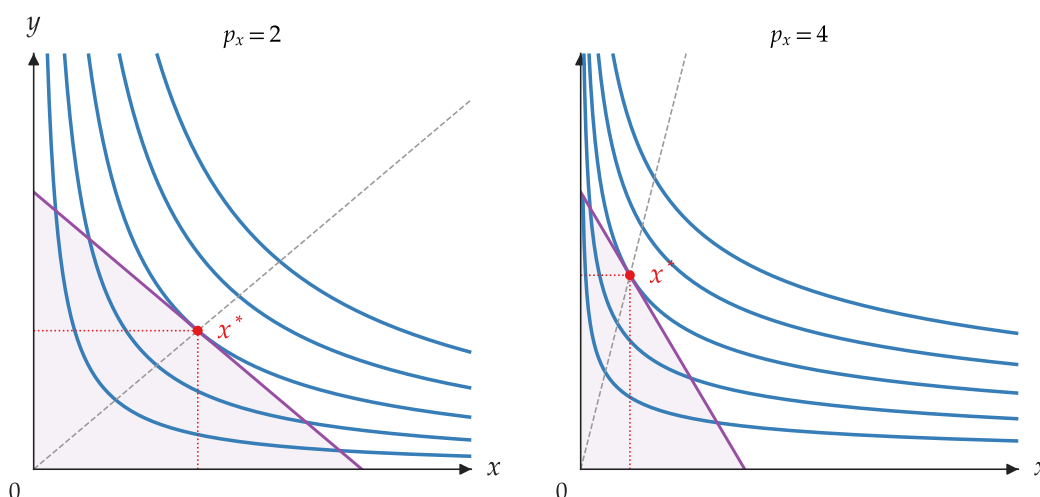
cases = [
 (CobbDouglas(alpha=0.5, beta=0.5), 2.0, 3.0, 30.0, r"Before: $p_x=2$"),
 (CobbDouglas(alpha=0.3, beta=0.7), 4.0, 3.0, 30.0, r"After: $p_x=4$"),
]

for idx, (model, px, py, income, title) in enumerate(cases):
 eq = solve(model, px=px, py=py, income=income)
 panel = fig[idx]
 panel.ax.set_title(title)
 panel.add_utility(model, levels=levels.around(eq.utility, n=5), label="IC")
 panel.add_budget(px, py, income, fill=True, label="BC")
 panel.add_equilibrium(eq, show_ray=True)

fig[0].show_legend(legend=Legend(position="upper right"))
fig.save("figure_side_by_side.png")

```

**Layout** The available layouts: `SINGLE`, `STACKED`, `SIDE_BY_SIDE`, `TOP_TWO_BOTTOM_ONE`, `TOP_ONE_BOTTOM_TWO`, `GRID_2X2` and `GRID_3X3`.  
*New: v1.2.0*



**Figure 11**  
Price change, before and after.

## 5.2 Paths

```
PricePath from econ_viz import IncomePath, LinearBudget, PricePath
IncomePath from econ_viz.models import CobbDouglas
```

```
model = CobbDouglas(alpha=0.5, beta=0.5)
budget = LinearBudget(px=2.0, py=2.0, income=40.0)
```

```
price_path = PricePath(
 model,
 budget=budget,
 price="px",
 price_range=(0.8, 6.0),
 n=40,
)
income_path = IncomePath(
 model,
 budget=budget,
 income_range=(20.0, 80.0),
 n=30,
)
```

*New: v1.2.0* Sweep one budget parameter while repeatedly solving the consumer's problem. Use a path to draw price- and income-consumption curves with `Canvas.add_path`, to feed a Demand Diagram, or to inspect how bundles move as prices or income vary.

```
Canvas.add_path from econ_viz import Canvas, levels

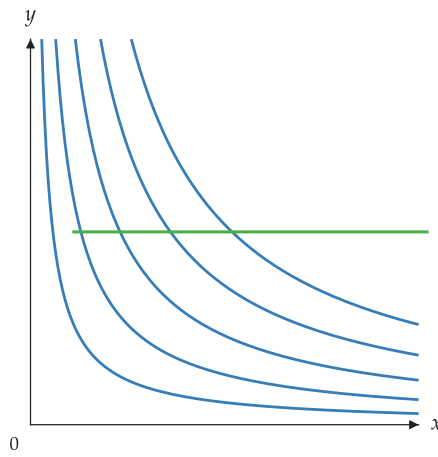
eq = price_path.equilibria[len(price_path.equilibria) // 2]
lvls = levels.around(eq.utility, n=5)

Canvas(x_max=25, y_max=20) \
 .add_utility(model, levels=lvls) \
 .add_path(price_path, label="PCC") \
 .save("price_path.png")
```

*New: v1.2.0* Draw a path directly on a canvas when a full demand diagram is not needed. For Cobb-Douglas preferences the price-consumption curve is horizontal (Figure 12): spending on  $y$  does not depend on  $p_x$ .

## 5.3 Demand diagrams

```
DemandDiagram DemandDiagram(<price path>, title=None, ...)
.add_marshallian_panel(price_markers=None, show_pcc=False,
show_demand_guides=True)
```



**Figure 12**  
The price-consumption curve as  $p_x$  varies.

New: v1.2.0

A stacked two-panel figure: indifference curves, budget lines and equilibria in the top panel, the corresponding Marshallian demand curve in the bottom one (Figure 13).

- DemandDiagram expects a PricePath.
- Smooth, kinked and corner demand are handled differently, so that the bottom panel stays economically meaningful.
- `show_pcc=True` overlays the price-consumption curve in goods space.

### Example 3

```
from econ_viz import DemandDiagram, LinearBudget, PricePath
from econ_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.5, beta=0.5)
budget = LinearBudget(px=2.0, py=2.0, income=40.0)
path = PricePath(
 model,
 budget=budget,
 price="px",
 price_range=(0.8, 6.0),
 n=40,
)

fig = DemandDiagram(path, title="Demand: Cobb-Douglas")
fig.add_marshallian_panel(
 price_markers=[1.5, 4.0],
 show_pcc=False,
 show_demand_guides=True,
)
fig.save("demand_cobb_douglas.png")
```

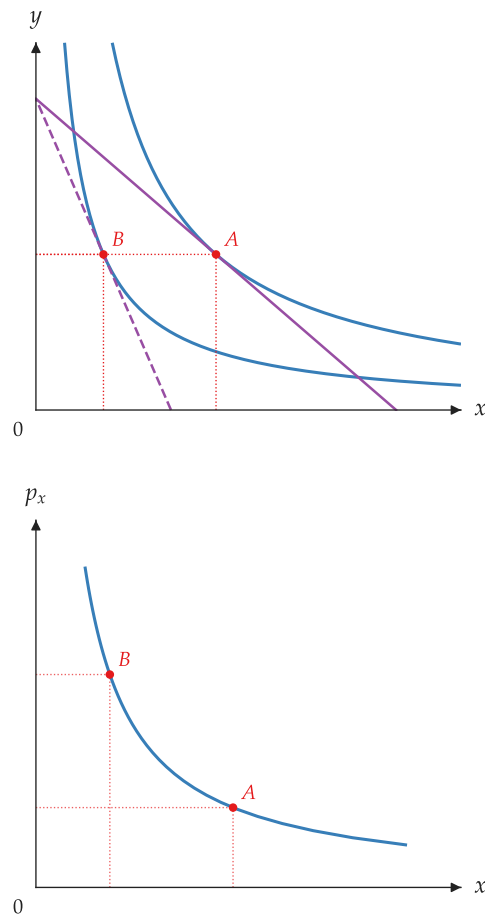
## 5.4 Price-effect decomposition

`decompose_price_effect` `decompose_price_effect((model), px=(old), (new)), py=(float), income=(float), method=(method))`

New: v1.5.0

Separate a price change into substitution and income effects. Choose `DecompositionMethod.HICKS` to hold the original utility fixed (Hicks, 1939), or `SLUTSKY` (Slutsky, 1915) to keep the original bundle affordable. The result has these fields:

|                                  |                                              |
|----------------------------------|----------------------------------------------|
| A, B, C                          | The original, compensated and final bundles. |
| <code>substitution_effect</code> | Substitution effect.                         |
| <code>income_effect</code>       | Income effect.                               |
| <code>total_effect</code>        | Total effect.                                |
| <code>compensated_income</code>  | Income after compensation.                   |

**Figure 13**

Demand curve at the optima.

|                              |                                                                                                                                                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Canvas.add_decomposition     | Draw the three bundles A, B and C, the budget lines, the indifference curves and the effect arrows. A Hicks decomposition draws the curves through A and C; a Slutsky decomposition also draws the one through B. |
| New: v1.5.0                  |                                                                                                                                                                                                                   |
| Updated: v1.9.0              |                                                                                                                                                                                                                   |
| show_curves                  | Draw the decomposition's indifference curves; set to False when you draw your own.                                                                                                                                |
| bool · default True          |                                                                                                                                                                                                                   |
| curve_stroke                 | Style of the indifference curves.                                                                                                                                                                                 |
| Stroke   None · default None |                                                                                                                                                                                                                   |
| curve_label                  | Show and style the $U_0$ , $U_1$ and $U_B$ labels.                                                                                                                                                                |
| Label   None · default None  |                                                                                                                                                                                                                   |
| substitution, income         | Style of the substitution and income effects (see Effect below).                                                                                                                                                  |
| Effect   None · default None |                                                                                                                                                                                                                   |
| point_marker                 | Markers of the bundles.                                                                                                                                                                                           |
| Marker   None · default None |                                                                                                                                                                                                                   |
| point_label                  | Text of the bundles.                                                                                                                                                                                              |
| Label   None · default None  |                                                                                                                                                                                                                   |
| legend                       | Position and style of the legend (see Legend below).                                                                                                                                                              |
| Legend   None · default None |                                                                                                                                                                                                                   |
| Effect                       | Effect(color=None, y=None, label=None, label_position="right", label_offset=4, opacity=None)                                                                                                                      |
| New: v1.8.0                  |                                                                                                                                                                                                                   |
| Updated: v1.9.0              | Colour, height of the arrow below the horizontal axis, label and opacity of the substitution or income effect. The line itself can still be overridden with substitution_stroke or income_stroke.                 |

Legend Legend(position="auto", fontsize=None, frame=None, columns=None, visible=None, opacity=None)

New: v1.9.0 "auto" picks the inner corner that hides the least of the plot; when all four corners overlap it, the legend moves to the right. Inner corners such as "upper left" and outer positions "top", "bottom", "left" and "right" can also be given. Legend(visible=False) hides the legend.

#### Example 4

```
from econ_viz import Canvas, DecompositionMethod, Effect, Label, Legend
from econ_viz.models import CobbDouglas
from econ_viz.optimizer import decompose_price_effect

model = CobbDouglas(alpha=0.5, beta=0.5)
result = decompose_price_effect(
 model,
 px=(2.0, 4.0),
 py=3.0,
 income=60.0,
 method=DecompositionMethod.HICKS,
)

(
 Canvas(x_max=25, y_max=25, title="Hicks decomposition")
 .add_decomposition(
 result,
 show_arrows=True,
 label_effects=True,
 show_x_projections=True,
 substitution=Effect(label="SE"),
 income=Effect(label="IE", opacity=0.8),
 curve_label=Label(position="right"),
 legend=Legend(position="bottom"),
)
 .save("hicks.png")
)
```

## 5.5 Edgeworth box

EdgeworthBox EdgeworthBox(<utility A>, <utility B>, total\_x=<float>, total\_y=<float>, ...)

New: v1.3.0 A two-consumer exchange economy (Edgeworth, 1881). Consumer A is measured from the lower-left origin and consumer B from the upper-right origin. Layers are added with add\_endowment, add\_contract\_curve, add\_core, add\_price\_line and add\_walrasian\_equilibrium.

Use method="mrs" for the contract curve of smooth preferences, and method="pareto" for models with kinks, corners or custom piecewise utility functions.

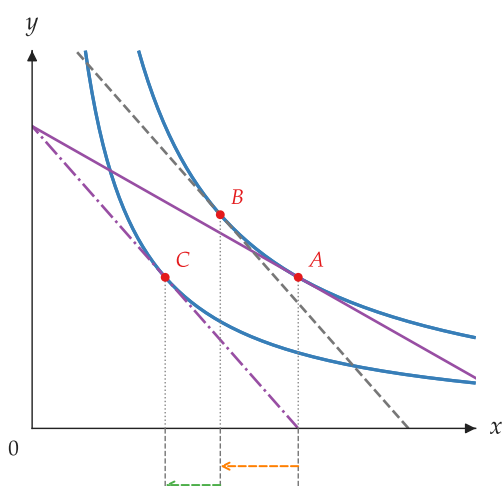
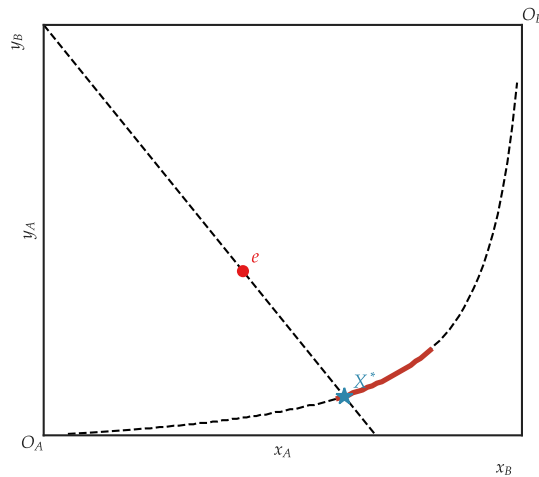


Figure 14  
Hicks decomposition.



**Figure 15**  
Contract curve and equilibrium.

### Example 5

```
from econ_viz import EdgeworthBox
from econ_viz.models import CobbDouglas

box = EdgeworthBox(
 CobbDouglas(alpha=0.8, beta=0.2),
 CobbDouglas(alpha=0.2, beta=0.8),
 total_x=12.0,
 total_y=10.0,
 title="Asymmetric Cobb-Douglas",
)

(
 box.add_endowment(5.0, 4.0)
 .add_contract_curve(n=100, method="mrs")
 .add_core()
 .add_price_line(px=1.2, py=1.0)
 .add_walrasian_equilibrium(px=1.2, py=1.0)
 .show_legend(loc="center left", bbox_to_anchor=(1.02, 0.5))
 .save("edgeworth.png")
)
```

## 6 Core models

The core models cover smooth, kinked, linear, income-dependent, and reference-point preferences. They all live in `econ_viz.models`. This chapter gives each model's function, parameters, distinct behaviour, and figure.

### 6.1 Smooth preferences

These models produce smooth indifference curves and normally have an interior optimum when prices and income are positive.

#### Cobb-Douglas

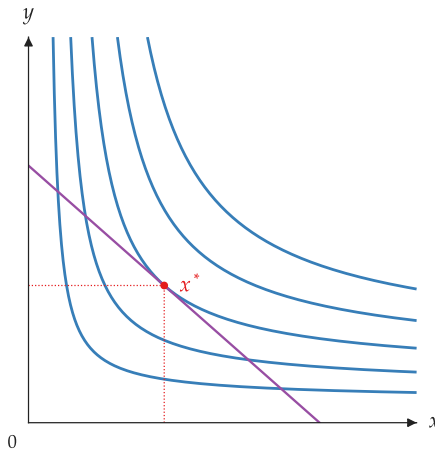
CobbDouglas CobbDouglas(alpha=0.5, beta=0.5)

The standard model for smooth, strictly convex preferences (Cobb & Douglas, 1928):

$$u(x, y) = x^\alpha y^\beta.$$

The exponents control the relative weight placed on each good. The indifference curves approach both axes without touching them.

alpha Weight on good  $x$ .  
default 0.5



**Figure 16**  
CES with  $\rho = -0.5$ .

beta Weight on good  $y$ .  
default 0.5

### CES

CES CES(alpha=0.5, beta=0.5, rho=0.5)

Constant elasticity of substitution (Arrow et al., 1961): the ease of substitution varies while preferences stay smooth,

$$u(x, y) = (\alpha x^\rho + \beta y^\rho)^{1/\rho}.$$

The elasticity of substitution is  $\sigma = 1/(1 - \rho)$ . As  $\rho$  changes, CES approaches Cobb-Douglas ( $\rho \rightarrow 0$ ), Leontief ( $\rho \rightarrow -\infty$ ) or perfect substitutes ( $\rho \rightarrow 1$ ).

alpha Weight on good  $x$ .  
default 0.5

beta Weight on good  $y$ .  
default 0.5

rho Substitution parameter,  $\rho \neq 1$ .  
default 0.5

### Translog

Translog Translog(alpha\_0=0.0, alpha\_x=0.5, alpha\_y=0.5, beta\_xx=0.0, beta\_yy=0.0, beta\_xy=0.0)

*New: v1.1.0* A flexible log-quadratic model for smooth preferences (Christensen et al., 1975):

$$\begin{aligned} \ln u(x, y) = & \alpha_0 + \alpha_x \ln x + \alpha_y \ln y \\ & + \frac{1}{2} \beta_{xx} (\ln x)^2 + \frac{1}{2} \beta_{yy} (\ln y)^2 + \beta_{xy} \ln x \ln y. \end{aligned}$$

The quadratic and interaction terms let the curvature vary across the consumption space. With all  $\beta$  coefficients zero it reduces to a Cobb-Douglas-style log-linear form.

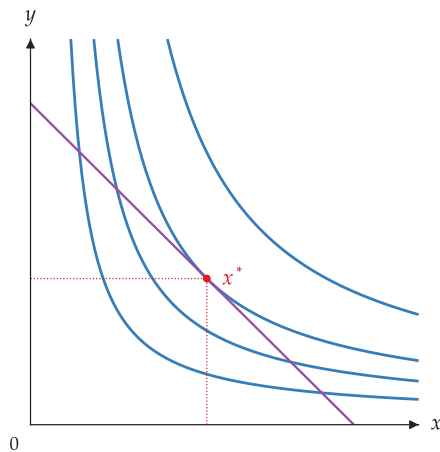
alpha\_0 Log-utility intercept.  
default 0.0

alpha\_x First-order weight on  $\ln x$ .  
default 0.5

alpha\_y First-order weight on  $\ln y$ .  
default 0.5

beta\_xx Curvature in  $x$ .  
default 0.0

beta\_yy Curvature in  $y$ .  
default 0.0



**Figure 17**  
Translog with  $\beta_{xy} = 0.12$ .

beta\_xy Interaction between  $x$  and  $y$ .  
default 0.0

## 6.2 Kinks and corners

These models show how non-smooth or linear preferences move the optimum to a kink or a corner.

### Perfect complements

Leontief Leontief(a=1.0, b=1.0)

Goods consumed in fixed proportions:

$$u(x, y) = \min(ax, by).$$

The indifference curves are L-shaped, and the optimum lies at the kink where the two weighted quantities are equal.

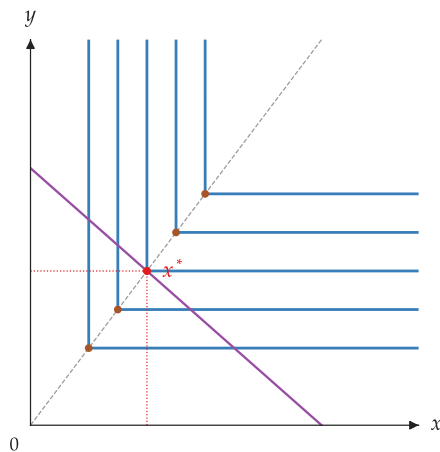
a Weight on good  $x$ .  
default 1.0

b Weight on good  $y$ .  
default 1.0

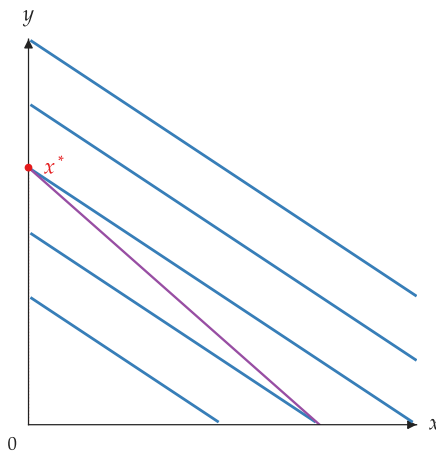
### Perfect substitutes

PerfectSubstitutes PerfectSubstitutes(a=1.0, b=1.0)

A constant rate of trade-off between the goods:



**Figure 18**  
Leontief, with the ray through the kinks.

**Figure 19**

Perfect substitutes: a corner solution on the  $y$ -axis.

$$u(x, y) = ax + by.$$

The indifference curves are straight lines. The consumer normally buys only the good with the greater marginal utility per dollar, a corner solution.

- a Marginal utility of good  $x$ .  
default 1.0
- b Marginal utility of good  $y$ .  
default 1.0

### Maximum utility

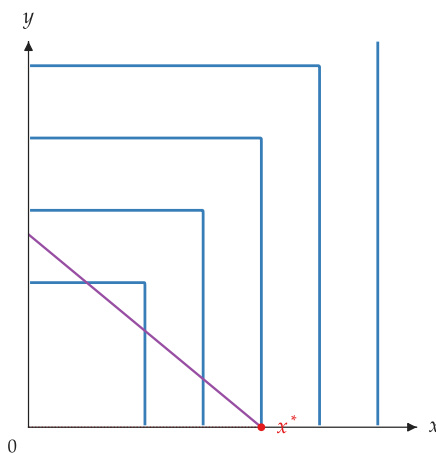
```
maximum CustomUtility(func=lambda x, y: np.maximum(a * x, b * y))
```

Maximum utility keeps only the larger weighted quantity in each bundle:

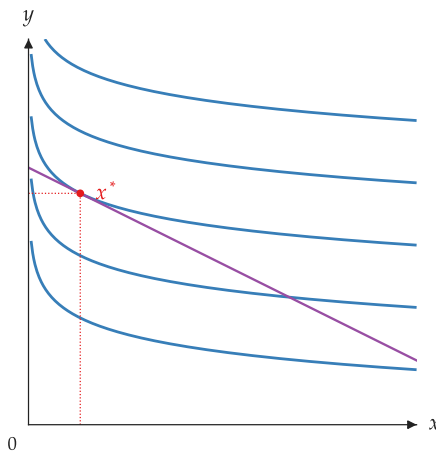
$$u(x, y) = \max(ax, by).$$

Each indifference curve has two arms that extend towards the axes. The preferences are non-convex, so on a linear budget the optimum normally lies at the intercept with the greater weighted utility. There is no dedicated class; the model is built with `CustomUtility` (Section 7).

- a Weight on good  $x$ .  
default 1.0
- b Weight on good  $y$ .  
default 1.0

**Figure 20**

Maximum utility.



**Figure 21**  
Quasi-linear preferences:  
vertically parallel indifference  
curves.

### 6.3 Income and reference points

These models add special income effects, subsistence requirements or a preferred consumption point.

#### Quasi-linear

QuasiLinear QuasiLinear(v\_func=numpy.log, linear\_in="y")

One good enters utility linearly:

$$u(x, y) = f(x) + y.$$

Once the solution is interior, the non-linear good has no income effect. `linear_in` reverses the roles of  $x$  and  $y$ .

v\_func Increasing, concave function  $f$ .  
default numpy.log  
linear\_in Good that enters linearly.  
default "y"

#### Haagsma

Haagsma Haagsma(alpha\_x=1.0, alpha\_y=2.0, gamma\_x=2.0, gamma\_y=27.0)

New: v1.9.0 The Haagsma (2012) utility function,

$$u(x, y) = \alpha_x \ln(x - \gamma_x) - \alpha_y \ln(\gamma_y - y).$$

It is defined for  $x > \gamma_x$  and  $0 \leq y < \gamma_y$ , and requires  $0 < \alpha_x < \alpha_y$ . Marshallian demand for  $x$  falls with income, and  $x$  is a Giffen good when  $\gamma_y p_y < I < \gamma_y p_y + \gamma_x p_x$ .

Once income reaches  $\gamma_y p_y + \gamma_x p_x$ , utility has no finite maximum on the budget line and the model raises `InvalidParameterError`.

alpha\_x Weight on good  $x$ ; must be less than `alpha_y`.  
default 1.0

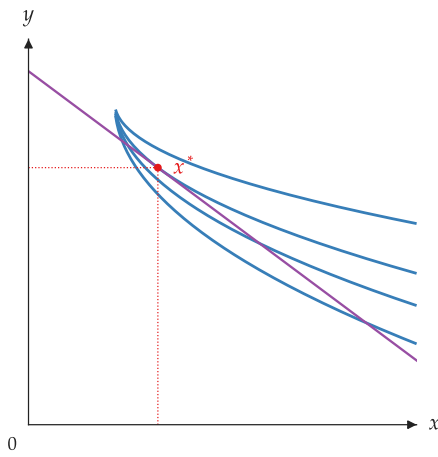
alpha\_y Weight on good  $y$ .  
default 2.0

gamma\_x Lower bound on the quantity of  $x$ .  
default 2.0

gamma\_y Upper bound on the quantity of  $y$ .  
default 27.0

demand(px, py, income) Closed-form interior solution.

is\_giffen(px, py, income) Whether  $x$  is a Giffen good at the given budget.



**Figure 22**  
Haagsma with  $\gamma_y = 10$ .

### Stone-Geary

StoneGeary StoneGeary(alpha=0.5, beta=0.5, bar\_x=1.0, bar\_y=1.0)

Cobb-Douglas with minimum consumption requirements (Geary, 1950; Stone, 1954):

$$u(x, y) = (x - \bar{x})^\alpha (y - \bar{y})^\beta.$$

The consumer first covers the subsistence quantities  $\bar{x}$  and  $\bar{y}$ , then allocates the remaining income like a Cobb-Douglas consumer.

alpha Weight on supernumerary  $x$ .  
default 0.5  
beta Weight on supernumerary  $y$ .  
default 0.5  
bar\_x Subsistence quantity  $\bar{x}$ .  
default 1.0  
bar\_y Subsistence quantity  $\bar{y}$ .  
default 1.0

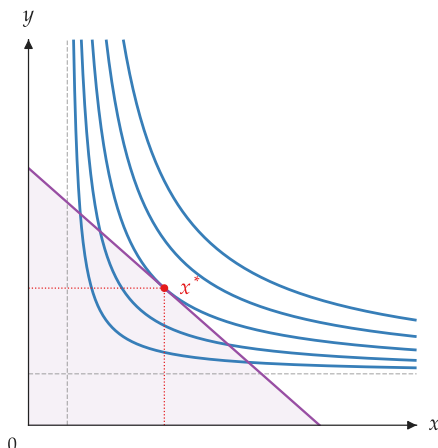
### Satiation

Satiation Satiation(bliss\_x=5.0, bliss\_y=5.0, a=1.0, b=1.0)

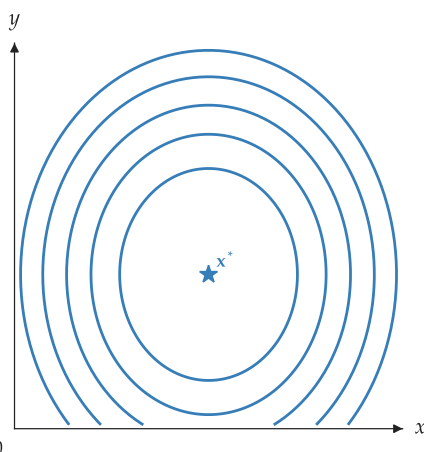
A bliss point that maximises utility:

$$u(x, y) = -a(x - x^*)^2 - b(y - y^*)^2.$$

Utility falls in every direction away from  $(x^*, y^*)$ , so the indifference curves are closed ellipses and monotonicity does not hold.



**Figure 23**  
Stone-Geary with  $\bar{x} = \bar{y} = 2$ .



**Figure 24**  
Satiation around the bliss point  
(6,4).

`bliss_x` Bliss-point coordinate  $x^*$ .  
default 5.0  
`bliss_y` Bliss-point coordinate  $y^*$ .  
default 5.0  
    a Curvature along the  $x$ -axis.  
default 1.0  
    b Curvature along the  $y$ -axis.  
default 1.0

## 7 Advanced models

Advanced models extend the built-in families with user-defined functions or more than two goods.

### Custom utility

`CustomUtility` `CustomUtility(func=<callable>, name=<str>)`

Wrap any vectorised Python callable as an `econ-viz` model. The callable must accept two NumPy arrays and return an array of the same shape. The example below uses

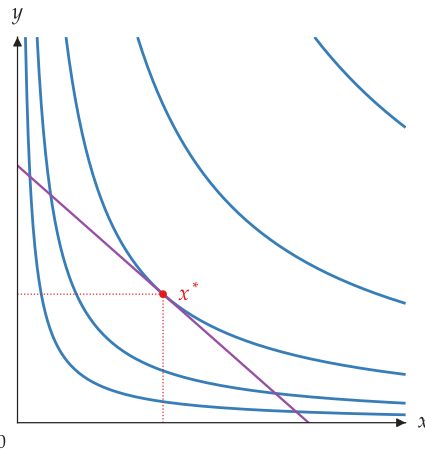
$$u(x, y) = \ln x + \ln y.$$

```
func Vectorised utility function of x and y .
name Display name of the model.

import numpy as np
from econ_viz import Canvas, levels, solve
from econ_viz.models import CustomUtility

model = CustomUtility(
 func=lambda x, y: np.log(x) + np.log(y),
 name="log+log",
)
eq = solve(model, px=2.0, py=3.0, income=30.0)

(
 Canvas(x_max=20, y_max=15, title="Custom Utility")
 .add_utility(model, levels=levels.around(eq.utility, n=5))
 .add_budget(2.0, 3.0, 30.0)
 .add_equilibrium(eq)
 .save("custom.png")
)
```



**Figure 25**  
A custom model,  $\ln x + \ln y$ .

### Many-good Cobb-Douglas

```
MultiGoodCD MultiGoodCD(<shares>)
MultiGoodCD.freeze .freeze(<good>=<quantity>, ...)
```

Cobb-Douglas preferences over  $N$  goods:

$$u(x_1, \dots, x_N) = \prod_{i=1}^N x_i^{\alpha_i}.$$

`freeze()` fixes every good except  $x$  and  $y$  and returns a `CustomUtility` that can be drawn on a two-dimensional canvas.

```
shares Mapping from each good's name to its exponent α_i .
freeze(...) Fixed quantities of the goods other than x and y .

from econ_viz import Canvas, levels, solve
from econ_viz.models import MultiGoodCD

model = MultiGoodCD({"x": 0.3, "y": 0.3, "z": 0.4})
two_good_model = model.freeze(z=10.0)
eq = solve(two_good_model, px=2.0, py=3.0, income=30.0)

(
 Canvas(x_max=20, y_max=15, title="Multi-Good Cobb-Douglas")
 .add_utility(
 two_good_model,
 levels=levels.around(eq.utility, n=5),
)
 .add_budget(2.0, 3.0, 30.0, fill=True)
 .add_equilibrium(eq)
 .save("multigood.png")
)
```

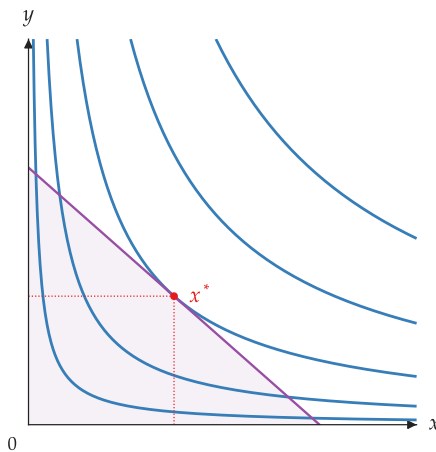
## 8 Analysis

The analysis API provides comparative statics of Marshallian demand, the Slutsky matrix, and checks for homogeneity and homotheticity of utility functions.

### 8.1 Comparative statics

```
comparative_statics comparative_statics(<model>, px=<float>, py=<float>, income=<float>)
```

*New: v1.1.0* Estimate the six partial derivatives of two-good Marshallian demand with respect to  $p_x, p_y$  and income by finite differences, returned as a `ComparativeStatics`.

**Figure 26**

A three-good Cobb-Douglas model with  $z = 10$  fixed.

- Central finite differences around `solve(...)`, with a default relative step of  $1e-3$ .
- Warns when the own-price derivative is positive or the income derivative negative, the signatures of Giffen and inferior goods.

`dx_dpx`  $\partial x^* / \partial p_x$ .  
float

`dx_dpy`  $\partial x^* / \partial p_y$ .  
float

`dx_dI`  $\partial x^* / \partial I$ .  
float

`dy_dpx`  $\partial y^* / \partial p_x$ .  
float

`dy_dpy`  $\partial y^* / \partial p_y$ .  
float

`dy_dI`  $\partial y^* / \partial I$ .  
float

```
from econ_viz.models import CobbDouglas
from econ_viz.optimizer import comparative_statics

model = CobbDouglas(alpha=0.4, beta=0.6)
cs = comparative_statics(model, px=2.0, py=3.0, income=60.0)

print(round(cs.dx_dpx, 1), round(cs.dx_dpy, 1), round(cs.dx_dI, 1))
print(round(cs.dy_dpx, 1), round(cs.dy_dpy, 1), round(cs.dy_dI, 1))

-6.0 0.0 0.2
0.0 -4.0 0.2
```

## 8.2 Slutsky matrix

`slutsky_matrix` `slutsky_matrix((model), px=(float), py=(float), income=(float))`

Updated: v1.2.3

The two-good Slutsky substitution matrix, computed from Marshallian demand derivatives and income effects. The result checks symmetry, negative semidefiniteness and homogeneity, and warns when a condition fails. `as_array()` returns it as a NumPy array.

```
from econ_viz import slutsky_matrix
from econ_viz.models import CobbDouglas

S = slutsky_matrix(
 CobbDouglas(alpha=0.4, beta=0.6),
 px=2.0, py=3.0, income=60.0,
)
```

```

print(round(S.s_xx, 1), round(S.s_xy, 1))
print(round(S.s_yx, 1), round(S.s_yy, 1))
print(S.as_array().round(1))

-3.6 2.4
2.4 -1.6
[[-3.6 2.4]
[2.4 -1.6]]

```

### 8.3 Homogeneity

HomogeneityAnalyzer HomogeneityAnalyzer(<model>)

*New: v1.1.0* Check homogeneity and homotheticity of a utility function, and degree-zero homogeneity of demand.

degree() Estimate the degree of homogeneity; see Section 8.4.

euler\_check(x, y) Euler-theorem residual at a bundle.

is\_homothetic() Whether the MRS is invariant to proportional scaling.

demand\_degree\_zero(px, py, income) Whether Marshallian demand is homogeneous of degree zero.

```

from econ_viz.analysis import HomogeneityAnalyzer
from econ_viz.models import CobbDouglas

analyzer = HomogeneityAnalyzer(CobbDouglas(alpha=0.4, beta=0.6))
result = analyzer.degree()

print(round(result.degree, 6))
print(result.returns_to_scale)
print(round(analyzer.euler_check(3.0, 4.0), 6))
print(analyzer.is_homothetic())
print(analyzer.demand_degree_zero(px=2.0, py=3.0, income=60.0))

1.0
ReturnsToScale.CONSTANT
0.0
True
True

```

### 8.4 Returns to scale

degree() returns a HomogeneityResult with these fields:

degree *float* Estimated degree of homogeneity.

is\_homogeneous *bool* Whether the function is homogeneous.

returns\_to\_scale *ReturnsToScale* Returns-to-scale class, one of the values below.

For Cobb-Douglas utility the degree is  $\alpha + \beta$ :

$$u(\lambda x, \lambda y) = \lambda^{\alpha+\beta} u(x, y).$$

INCREASING  $\alpha + \beta > 1$ , increasing returns.

CONSTANT  $\alpha + \beta = 1$ , constant returns.

DECREASING  $\alpha + \beta < 1$ , decreasing returns.

NOT\_HOMOGENEOUS No consistent degree of homogeneity.

```

def shifted_utility(x, y):
 return x**0.4 * y**0.6 + 1.0

```

```
models = [
```

```

CobbDouglas(alpha=0.7, beta=0.6),
CobbDouglas(alpha=0.4, beta=0.6),
CobbDouglas(alpha=0.2, beta=0.5),
shifted_utility,
]

for model in models:
 result = HomogeneityAnalyzer(model).degree()
 degree = None if result.degree is None else round(result.degree, 1)
 print(degree, result.returns_to_scale.name)

1.3 INCREASING
1.0 CONSTANT
0.7 DECREASING
None NOT_HOMOGENEOUS

```

## 9 Themes and configuration

A Theme defines the default look of a diagram; a Config loads theme overrides and font settings from a TOML file. Canvas, Figure, DemandDiagram, price-effect decompositions and EdgeworthBox share these style objects.

### 9.1 Built-in themes

```

themes from econ_viz import Canvas, themes

cvs = Canvas(theme=themes.paper)

```

*Updated: v1.11.0* The Python API provides the themes below. Theme objects are immutable; to change a field, create a new Theme or override the field with a Config.

**Table 4**  
Built-in themes

| Name         | Character                                                   |
|--------------|-------------------------------------------------------------|
| default      | Colour-blind-friendly default palette                       |
| colorblind   | Alias with the same palette as default                      |
| nord         | Cool theme on the Nord palette                              |
| paper        | Transparent background and fine lines for print             |
| monochrome   | Black and white; layers told apart by line style and marker |
| presentation | Larger labels, lines and markers for projection             |
| dark         | Dark background with matching foreground colours            |

The Python API and the base key of a configuration file accept all seven names<sup>5</sup>.

The default theme takes its colours from a colour-blind-friendly palette (thriveth, 2014). `themes.COLORBLIND_CYCLE_HEX` and `themes.COLORBLIND_CYCLE_RGB` give the palette as hexadecimal and RGB values. Economics teaching leans heavily on diagrams, so meaning should never rest on colour alone, or visually impaired students cannot follow it (Kugler & Andrews, 1996).

<sup>5</sup>Passed directly to `--theme` on the command line, only the six themes other than the `colorblind` alias are accepted.

## 9.2 Custom themes

```

Theme from econ_viz import Theme

my_theme = Theme(
 name="custom",
 background_color="white",
 label_scale=1.0,
 axis_color="#333333",
 label_color="#333333",
 ic_color="#2563eb",
 secondary_ic_color="#93c5fd",
 secondary_ic_opacity=0.45,
 budget_color="#dc2626",
 eq_color="#16a34a",
)

```

Updated: v1.12.0 A Theme holds default colours, widths and styles for diagram elements. Fields not given keep the class defaults; style arguments passed explicitly to a drawing method take precedence over the theme.

|                           |                                                               |
|---------------------------|---------------------------------------------------------------|
| name                      | Theme identifier.                                             |
| str                       |                                                               |
| background_color          | Background of the figure and axes; None keeps it transparent. |
| str   None · default None |                                                               |
| label_scale               | Multiplier for ordinary label sizes.                          |
| float · default 1.0       |                                                               |
| axis_color                | Axis colour.                                                  |
| str · default "#222222"   |                                                               |
| label_color               | Colour of axis labels and origin text.                        |
| str · default "#222222"   |                                                               |
| ic_color                  | Indifference-curve colour.                                    |
| str · default "#377EB8"   |                                                               |
| ic_linewidth              | Indifference-curve width.                                     |
| float · default 1.8       |                                                               |
| secondary_ic_color        | Colour of non-focal indifference curves; None uses ic_color.  |
| str   None · default None |                                                               |
| secondary_ic_linewidth    | Width of non-focal indifference curves.                       |
| float · default 1.0       |                                                               |
| secondary_ic_opacity      | Opacity of non-focal indifference curves.                     |
| float · default 0.45      |                                                               |
| path_color                | Colour of PCC and ICC paths.                                  |
| str · default "#4DAF4A"   |                                                               |
| budget_color              | Budget-line colour.                                           |
| str · default "#984EA3"   |                                                               |
| budget_fill_alpha         | Opacity of the budget-set shading.                            |
| float · default 0.08      |                                                               |
| eq_color                  | Colour of the equilibrium and its drop lines.                 |
| str · default "#E41A1C"   |                                                               |
| eq_markersize             | Equilibrium marker size.                                      |
| float · default 4.0       |                                                               |

Theme also holds colours and widths for rays, kinks, compensated budget lines and price-effect arrows. `axis_stroke`, `drop_stroke`, `projection_stroke`, `guide_stroke` and `box_stroke` hold the defaults of other lines; markers, labels, fills and legends come from `Marker`, `Label`, `Fill` and `Legend` properties (Section 4.3).

### 9.3 Configuration files

```
Config.load from econ_viz import Config
Config.use Config.load("econ-viz.toml").use()
Config.reset # diagrams created from here on use these settings

Config.reset() # back to the built-in defaults
```

*New: v1.10.0* `Config.load()` reads a TOML file and returns a `Config`; `use()` makes it the default for later diagrams, and `reset()` restores the built-in defaults.

#### Tutorial: one style for every diagram

Figures in one set of lecture notes or one paper usually share their colours and line styles. Rather than repeating the same arguments on every `Canvas`, write the style once in an `econ-viz.toml` at the project root, where both Python and the command line read it.

1. **Create a template.** Run the following in the project root. It writes a commented `econ-viz.toml` listing the names and fields each section accepts:

```
econ-viz init
```

Every setting in the template starts commented out; keep only what you change and delete the rest.

2. **Pick a base theme.** `base` names the built-in theme to start from (Table 4); everything else overrides it:

```
base = "paper"
```

3. **Override colours and styles.** Write only the fields you change:

```
[color]
ic = "#1B4F72"
eq = "#C0392B"

[stroke.budget]
width = 1.6
style = "dashed"

[marker.eq]
size = 5

[label.point]
fontsize = 11
position = "right"
```

Section names map to theme properties: `ic` under `[color]` sets `ic_color`, and `[stroke.budget]` sets `theme.budget_stroke`. The fields inside a section are the arguments of `Stroke`, `Marker`, `Label`, `Fill` or `Legend` (Section 4.3). Fonts go under `[font]` as `text` and `math`. Fields you leave out keep the base theme's values.

4. **Use it from Python.** Load it once, before creating any diagram:

```
from econ_viz import Config

Config.load().use() # reads ./econ-viz.toml by default
```

Every `Canvas`, `Figure`, `DemandDiagram` and `EdgeworthBox` created afterwards uses these settings; `Config.reset()` restores the defaults.

5. **Use it from the command line.** Pass the file with `--config`:

```
econ-viz plot --config econ-viz.toml --model cobb-douglas \
--px 2 --py 3 --income 30 --output figure.png
```

Settings apply in this order, highest first: arguments passed to `Canvas` or a drawing method, then the configuration file, then the base theme. For example, `Canvas(theme=themes.default)` ignores the file's theme.

A mistake in the file makes `Config.load()` raise `InvalidParameterError` with the valid names. For instance, writing `[stroke.budgt]` for `[stroke.budget]` gives:

```
econ-viz.toml: [stroke.budgt]: no such setting (choose: axis, box,
budget, ...)
```

## 10 Export

### 10.1 Output formats

```
Canvas.save cvs.save(<path>)
```

Write the figure to path; the format follows the extension. Figure, DemandDiagram and EdgeworthBox have the same method.

- PNG: raster, at the canvas dpi (default 300).
- PDF, SVG: vector output that stays sharp when printed or scaled.
- TeX: TikZ code (.tex) for inclusion in L<sup>A</sup>T<sub>E</sub>X documents; adjust it with the keyword arguments `tikz_scale` and `tikz_standalone`.

```
cvs.save("figure.png") # PNG, canvas DPI (default 300)
cvs.save("figure.pdf") # PDF (vector)
cvs.save("figure.svg") # SVG (vector)
cvs.save("figure.tex") # TikZ
```

The dpi parameter of Canvas only affects raster output; lower it for faster previews:

```
cvs = Canvas(x_max=20, y_max=15, dpi=150) # lower DPI, faster preview
```

### 10.2 Animated GIFs

Export a GIF with `Animator.save()`; Section 12 has complete examples.

```
from econ_viz.animation import Animator
```

```
Animator(draw_frame, frames=frames).save("animation.gif", fps=12, dpi=120)
```

### 10.3 Interactive window

`cvs.show()` opens a live `matplotlib` window instead of saving. In the CLI, omitting `--output` has the same effect (Section 11.2):

```
econ-viz plot --model cobb-douglas --px 2 --py 3 --income 30
```

## 11 Command-line interface

The command-line interface plots diagrams, runs batches and prints demand formulas. Install it as a standalone tool with `uv tool install econ-viz`, or prefix each command with `uv run` inside a uv project (Section 2.5).

**Table 5**  
CLI commands

| Command                                      | Description                                                 |
|----------------------------------------------|-------------------------------------------------------------|
| <code>econ-viz help [&lt;command&gt;]</code> | Show help for the CLI or a specific command                 |
| <code>econ-viz models</code>                 | List all supported utility models                           |
| <code>econ-viz plot ...</code>               | Generate and export a diagram                               |
| <code>econ-viz solve-tex ...</code>          | Print a closed-form Marshallian demand as TeX               |
| <code>econ-viz init [path]</code>            | Create an <code>econ-viz.toml</code> configuration template |

## 11.1 Help and models

`econ-viz help` `econ-viz help [<command>]`

Without an argument, list all commands; with one, show its options.

```
econ-viz help # all commands
econ-viz help plot # plot options
econ-viz help models # models options
```

`econ-viz models` List the model names and parameters the CLI supports. `--model` takes kebab-case names, which differ from the Python class names (Table 6). The parameter options are described in Section 11.2.

**Table 6**  
CLI models

| Model               | --model value       | Required         | Optional                                             |
|---------------------|---------------------|------------------|------------------------------------------------------|
| Cobb-Douglas        | cobb-douglas        | alpha, beta      | –                                                    |
| CES                 | ces                 | rho              | alpha, beta                                          |
| Perfect complements | leontief            | a, b             | –                                                    |
| Perfect substitutes | perfect-substitutes | a, b             | –                                                    |
| Satiation           | satiation           | bliss-x, bliss-y | a, b                                                 |
| Quasi-linear        | quasi-linear        | –                | v-func, linear-in                                    |
| Stone-Geary         | stone-geary         | –                | alpha, beta, bar-x, bar-y                            |
| Translog            | translog            | –                | alpha-0, alpha-x, alpha-y, beta-xx, beta-yy, beta-xy |

## 11.2 Plotting

`econ-viz plot` `econ-viz plot --model <name> <options>`  
`econ-viz plot --latex <expression> <options>`

Give the model with `--model`, or the utility function with `--latex`, not both. With `--output` the diagram is written to a file; without it an interactive window opens.

The budget line and equilibrium need both prices and income. `--no-budget`, `--no-equilibrium` and `--no-curves` each omit one layer and can be combined (Table 7).

**Table 7**  
Layer options

| Case                                                                      | Result                                                   |
|---------------------------------------------------------------------------|----------------------------------------------------------|
| <code>--px</code> , <code>--py</code> and <code>--income</code> all given | Budget line drawn and equilibrium solved                 |
| Any price or income missing                                               | Indifference curves only, no budget line or equilibrium  |
| <code>--no-equilibrium</code> added                                       | No solve and no equilibrium, even with prices and income |
| <code>--no-budget</code> added                                            | Budget line hidden, even with prices and income          |
| <code>--no-curves</code> with prices and income                           | Budget line and equilibrium only, no indifference curves |

```
named model
econ-viz plot --model cobb-douglas --alpha 0.5 --beta 0.5 ...

LaTeX expression
econ-viz plot --latex "x^{0.4} y^{0.6}" ...
```

### Example 6

```
Cobb-Douglas, shaded budget set
econ-viz plot --model cobb-douglas --alpha 0.5 --beta 0.5 \
 --px 2 --py 3 --income 30 \
 --fill --output cobb_douglas.png

LaTeX input, Nord theme, expansion path
econ-viz plot --latex "x^{0.4} y^{0.6}" \
 --px 2 --py 3 --income 30 \
 --theme nord --show-ray \
 --output cd_latex.png

perfect complements, larger canvas
econ-viz plot --model leontief --a 1 --b 2 \
 --px 2 --py 3 --income 30 \
 --x-max 20 --y-max 15 \
 --output leontief.png

CES, curves only
econ-viz plot --model ces --rho -0.5 \
 --x-max 20 --y-max 15 --n-curves 6 \
 --no-budget --no-equilibrium \
 --output ces.png

satiation (bliss point)
econ-viz plot --model satiation --bliss-x 6 --bliss-y 4 \
 --x-max 12 --y-max 10 \
 --no-budget --no-equilibrium \
 --output satiation.png

no --output: open a window
econ-viz plot --model cobb-douglas --px 2 --py 3 --income 30
```

### Model selection

- model, -m Model name; econ-viz models lists them all.
- latex, -l L<sup>A</sup>T<sub>E</sub>X expression for Cobb-Douglas, perfect complements, perfect substitutes or CES (Section 14).

### Prices and income

- px Price of good  $x$ .
- py Price of good  $y$ .
- income Consumer income.

### Model parameters

- alpha Alpha parameter (Cobb-Douglas, CES).  
default 0.5
- beta Beta parameter (Cobb-Douglas, CES).  
default 0.5
- a Parameter  $a$  (perfect complements, perfect substitutes, satiation).  
default 1.0
- b Parameter  $b$  (perfect complements, perfect substitutes, satiation).  
default 1.0
- rho Substitution parameter (CES).  
default 0.5
- bliss-x Bliss-point  $x$ -coordinate (satiation).  
default 5.0
- bliss-y Bliss-point  $y$ -coordinate (satiation).  
default 5.0

Quasi-linear, Stone-Geary and Translog take these further options:

`--v-func` Non-linear function of the quasi-linear model, log or sqrt.  
 default log  
`--linear-in` Good that enters the quasi-linear model linearly, x or y.  
 default y  
`--bar-x` Stone-Geary subsistence quantity  $\bar{x}$ .  
 default 1.0  
`--bar-y` Stone-Geary subsistence quantity  $\bar{y}$ .  
 default 1.0  
`--alpha-0` Translog intercept.  
 default 0.0  
`--alpha-x` Translog first-order weight on  $\ln x$ .  
 default 0.5  
`--alpha-y` Translog first-order weight on  $\ln y$ .  
 default 0.5  
`--beta-xx` Translog curvature in  $x$ .  
 default 0.0  
`--beta-yy` Translog curvature in  $y$ .  
 default 0.0  
`--beta-xy` Translog interaction between  $x$  and  $y$ .  
 default 0.0

```
econ-viz plot --model stone-geary --bar-x 2 --bar-y 2 \
 --px 2 --py 3 --income 30 \
 --x-max 20 --y-max 15 --output stone_geary.png
```

Subsistence spending here is  $2 \times 2 + 3 \times 2 = 10$ , below the income of 30. With an income of 10 or less, the solve fails because the budget lies outside the model's domain.

### Canvas and layers

`--x-max` Horizontal axis limit.  
 default 10  
`--y-max` Vertical axis limit.  
 default 10  
`--x-label` Horizontal axis label.  
 default x  
`--y-label` Vertical axis label.  
 default y  
`--title` Figure title.  
`--theme` Theme name: default, nord, paper, monochrome, presentation or dark (Section 9).  
 default default  
`--config` Path to a TOML configuration file. Command-line options take precedence over the file.  
`--n-curves` Number of indifference curves.  
 default 5  
`--dpi` Raster output resolution.  
 default 300  
`--fill` Shade the feasible set below the budget line.  
`--show-ray` Draw the expansion-path ray through the optimum.  
`--no-budget` Omit the budget line.  
`--no-equilibrium` Omit the equilibrium point.  
`--no-curves` Omit the indifference curves.  
`--output, -o` Output file (.png, .pdf, .svg, .tex); omit it to open an interactive window.  
 For batch runs, give each parameter set its own file name and create the output directory first. `--x-max` and `--y-max` do not follow the prices; when an intercept or the optimum falls outside the canvas, raise the limits as well.

### 11.3 Creating a configuration file

`econ-viz init` `econ-viz init [<path>] [--force]`

*New: v1.10.0* Write a commented `econ-viz.toml` template. Without a path it goes in the current directory; an existing file is overwritten only with `--force`.

```
econ-viz init
econ-viz init config/figures.toml
econ-viz plot --config econ-viz.toml --model cobb-douglas \
 --px 2 --py 3 --income 30 --output figure.png
```

### 11.4 Demand formulas

`econ-viz solve-tex` `econ-viz solve-tex --model <name> <options>`

Print the closed-form Marshallian demand as TeX, for any document that supports TeX math.

```
numeric parameters
econ-viz solve-tex --model cobb-douglas --alpha 0.4 --beta 0.6

symbolic parameters
econ-viz solve-tex --model cobb-douglas --symbolic-params

custom price and income symbols
econ-viz solve-tex --model leontief --a 2 --b 3 \
 --px-symbol p_1 --py-symbol p_2 --income-symbol M
```

Supports cobb-douglas, leontief, perfect-substitutes and their L<sup>A</sup>T<sub>E</sub>X expressions. The command prints the formula only and writes no figure.

`--symbolic-params` Keep model parameters as symbols, such as  $\alpha$  and  $\beta$ ; without it the given values are substituted.

`--px-symbol` Symbol for the price of  $x$ .  
     default `p_x`

`--py-symbol` Symbol for the price of  $y$ .  
     default `p_y`

`--income-symbol` Symbol for income.  
     default `I`

## 12 Animation

Animator calls a drawing function over a sequence of parameter values and exports the returned Canvas or Figure objects as a GIF. It needs the animation extra (Section 2.4).

Animator `Animator(<draw>, frames=<values>).save(<path>, fps=12, dpi=120, loop=0)`

*New: v1.4.0* Call `draw(value)` for every value in frames and write the canvases as the frames of a GIF.

- `save()` uses Pillow, so no ffmpeg dependency is required.
- Frames are composited onto white before export, which prevents frame-stacking artefacts.
- `fps`, `dpi` and `loop` set the frame rate, resolution and loop count.

## Example 7

```

import numpy as np

from econ_viz import Canvas, levels, solve
from econ_viz.animation import Animator
from econ_viz.models import CobbDouglas

def draw(px: float) -> Canvas:
 model = CobbDouglas(alpha=0.5, beta=0.5)
 eq = solve(model, px=px, py=2.0, income=20.0)
 lvls = levels.around(eq.utility, n=5)

 return (
 Canvas(
 x_max=14, y_max=12,
 x_label="X_1", y_label="X_2",
 title="Price sweep"
)
 .add_utility(model, levels=lvls)
 .add_budget(px=px, py=2.0, income=20.0, fill=True)
 .add_equilibrium(eq, show_ray=True, drop_dashes=True)
)

Animator(draw, frames=np.linspace(1.0, 6.0, 45)).save(
 "price_sweep.gif",
 fps=12,
 dpi=120,
)

```

examples/scripts/animation.py provides four examples<sup>6</sup>:

- Parameter sweep: prices and income fixed, a utility-function parameter varies.
- Price sweep: utility function and  $p_y$  fixed,  $p_x$  varies and the budget line rotates, as in the example above.
- Income sweep: utility function and prices fixed, income varies and the budget line shifts.
- Budget-only sweep: the budget line alone, without the utility layers.

The four differ only in which values `draw()` holds fixed and which one it varies. For the parameter sweep, let the `draw()` above take `alpha` and fix the price at  $p_x = 2$ :

```

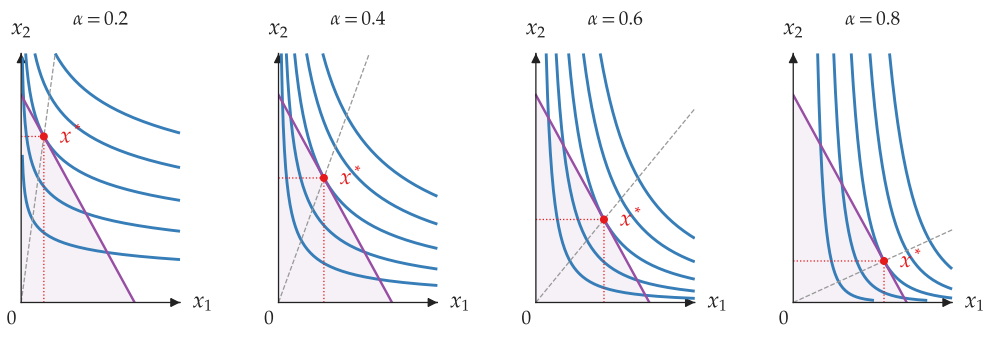
def draw(alpha: float) -> Canvas:
 model = CobbDouglas(alpha=alpha, beta=1.0 - alpha)
 eq = solve(model, px=2.0, py=2.0, income=20.0)
 ...

Animator(draw, frames=np.linspace(0.1, 0.9,
45)).save("parameter_sweep.gif")

```

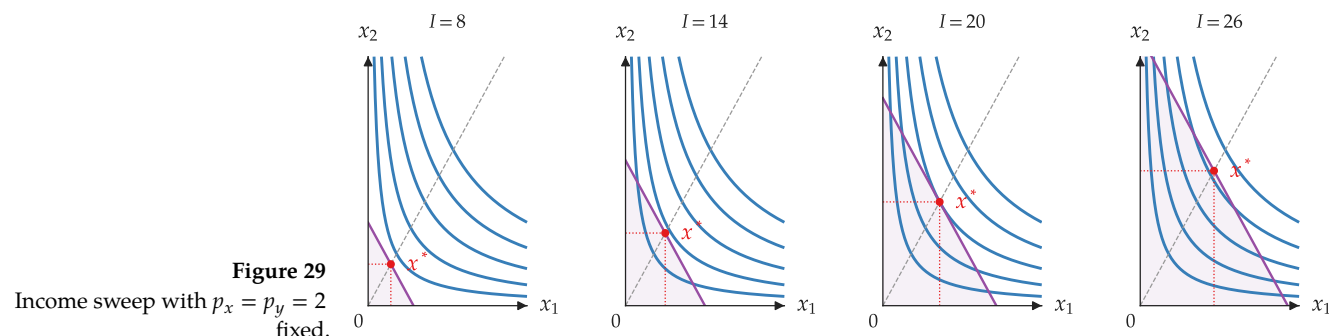
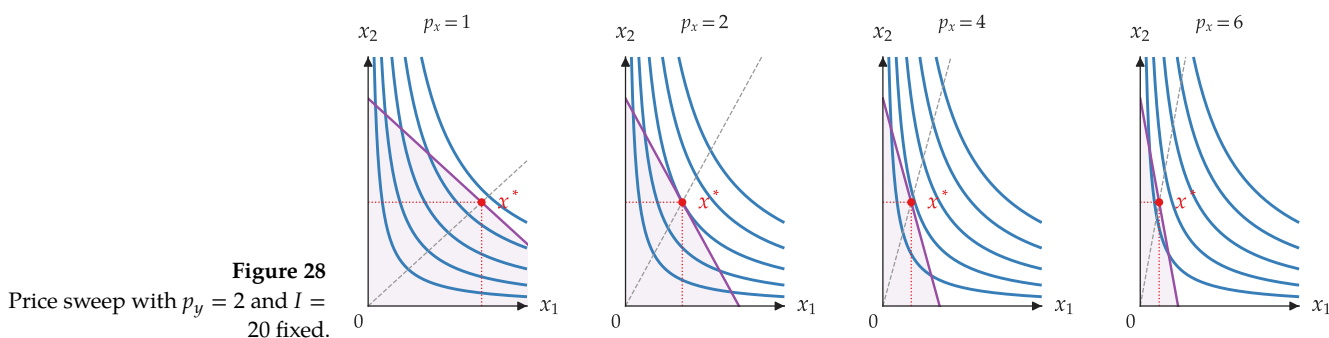
For the income sweep, `draw()` takes income and passes it to `add_budget()` and `solve()`. For the budget line alone, leave out `add_utility()` and `add_equilibrium()`.

Figure 27 to Figure 29 each show four frames.



**Figure 27**  
Parameter sweep:  $\alpha$  varies with  
 $\beta = 1 - \alpha$ .

<sup>6</sup>Each example covers Cobb-Douglas, CES, perfect substitutes and perfect complements.



## 13 Interactive widgets

WidgetViewer provides sliders and numeric input boxes in Jupyter and redraws the figure when a parameter changes. It needs the interactive extra (Section 2.4).

WidgetViewer WidgetViewer( $\langle draw \rangle$ ,  $\langle name \rangle = (\langle min \rangle, \langle max \rangle, \langle step \rangle)$ , ...).show()

New: v1.4.0

Pass a drawing function `draw` and give each parameter's range as  $(min, max, step)$ . Each parameter gets a linked `FloatSlider` and `FloatText`; when a value changes, `draw` is called again and its figure replaces the old one in the cell.

`draw` follows the same contract as for `Animator` (Section 12), but may take several parameters. Starting from the price-sweep example there, let `draw()` also take `alpha`:

### Example 8

```
from econ_viz.interactive import WidgetViewer

def draw(alpha: float, px: float) -> Canvas:
 model = CobbDouglas(alpha=alpha, beta=1.0 - alpha)
 ... # rest as in the animation example

WidgetViewer(
 draw,
 alpha=(0.2, 0.8, 0.05),
 px=(1.0, 6.0, 0.25),
).show()
```

### 13.1 Notebook setup

In a fresh Jupyter or Colab runtime, run the steps in this order<sup>7</sup>:

1. Run the install cell.
2. If the install upgraded `ipywidgets`, `traitlets` or `IPython`, restart the runtime.
3. After the restart, continue from the import cell without reinstalling.

<sup>7</sup>The Playground notebook in the `econ-viz` repository, `notebook/econ-viz Playground.ipynb`, follows this flow; open it in Jupyter, VS Code or Colab. It does not reinstall when `econ-viz` is already present.

13.2 Widgets or GIFs

Use `WidgetViewer` when users should adjust the parameters themselves; use `Animator` to export a GIF when a fixed sweep should play in slides or on a web page (Section 12).

14 L<sup>A</sup>T<sub>E</sub>X parsing

```
parse_latex parse_latex(<expression>)
Parse a LATEX utility function into a model, usable with solve(), Canvas or any other API that takes a model.
from econ_viz import parse_latex

model = parse_latex(r"x^{0.4} y^{0.6}") # CobbDouglas(alpha=0.4,
beta=0.6)
model = parse_latex(r"\min(2x, 3y)") # Leontief(a=2.0, b=3.0)
model = parse_latex(r"2x + 3y") # PerfectSubstitutes(a=2.0,
b=3.0)
```

14.1 Supported forms

Table 8 lists the accepted forms. Omitted coefficients and exponents default to 1.

Table 8  
Utility functions recognised by  
`parse_latex`.

| Family              | Pattern                                          | Example                            |
|---------------------|--------------------------------------------------|------------------------------------|
| Cobb-Douglas        | $x^{\alpha} y^{\beta}$ or $x^{\alpha} y^{\beta}$ | $x^{0.3} y^{0.7}$                  |
| Perfect complements | $\min(ax, by)$ or $\min(ax, by)$                 | $\min(2x, y)$                      |
| Perfect substitutes | $ax + by$                                        | $3x + 1.5y$                        |
| CES                 | $(\alpha x^{\rho} + \beta y^{\rho})^{1/\rho}$    | $(0.4x^{-0.5} + 0.6y^{-0.5})^{-2}$ |

A leading  $U =$  or  $u(x, y) =$  may be omitted:  
 $U(x,y) = x^{0.5} y^{0.5}$   
 $U = x^{0.5} y^{0.5}$   
 $x^{0.5} y^{0.5}$

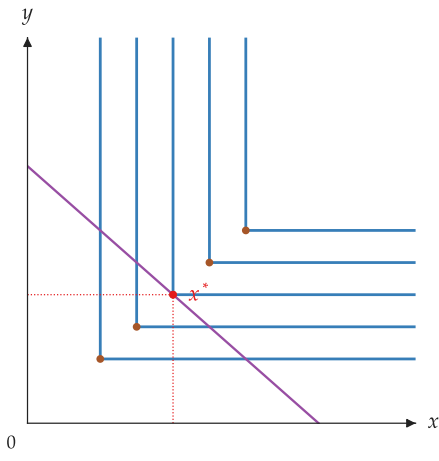
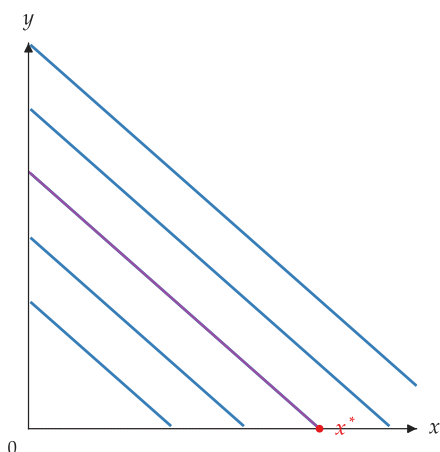


Figure 30  
`parse_latex(r"\min(2x, 3y)").`



**Figure 31**  
`parse_latex(r"2x + 3y").`

## 14.2 Errors

Unparseable input raises `econ_viz.exceptions.ParseError`, whose message lists the supported forms:

```
from econ_viz import parse_latex
from econ_viz.exceptions import ParseError

try:
 model = parse_latex(r"x^2 + y^2")
except ParseError as e:
 print(e)

Unrecognised LaTeX utility function: 'x^2 + y^2'
Supported forms:
Cobb-Douglas : x^{alpha} y^{beta}
Leontief : \min(ax, by)
Perfect Substitutes: ax + by
CES : (alpha x^{rho} + beta y^{rho})^{1/rho}
```

## 14.3 In the command-line tool

`econ-viz plot --latex` uses the same parser and accepts the same input (Section 11.2):

```
econ-viz plot --latex "x^{0.4} y^{0.6}" --px 2 --py 3 --income 30 -o
out.png
```

## Change history

Release history of econ-viz. Documentation-only changes are not listed; each entry is tagged where the feature it describes is documented, following l3doc convention, so the page number is the real page.

|                                                                                                                                                      |              |                                                                                                                                              |              |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>v1.12.0</b>                                                                                                                                       | (2026-09-26) | DemandDiagram: Supports Legend, Marker, Label and per-layer Stroke                                                                           | 15           |
| <b>General:</b> Focal and secondary indifference-curve levels, with numeric or ordinal labels                                                        | 2            |                                                                                                                                              |              |
| Canvas.add_utility: Focal utility level, secondary curve style and ordinal labels                                                                    | 7            | Canvas.add_decomposition: Draws the indifference curves through A, B and C by default, with curve labels and Legend                          | 16           |
| Theme: Colour, width and opacity of secondary indifference curves                                                                                    | 29           | Effect arrows below the horizontal axis point A→B and B→C; zero effects draw no arrow                                                        | 16           |
| <b>v1.11.0</b>                                                                                                                                       | (2026-09-26) | Haagsma: Haagsma utility model with a closed-form solution, for inferior- and Giffen-good analysis                                           | 23           |
| <b>General:</b> Complete theme system: one visual language across canvases, multi-panel figures, demand diagrams, decompositions and Edgeworth boxes | 2            | <b>v1.8.0</b>                                                                                                                                | (2026-09-26) |
| Canvas: Backgrounds, labels, lines and markers all take their defaults from the theme                                                                | 7            | Canvas: Axis, Marker, Label and Fill style objects                                                                                           | 7            |
| Figure: Panel backgrounds, labels, lines and markers fully follow the theme                                                                          | 14           | Stroke: Every line style is a Stroke; the individual style arguments remain as shorthands                                                    | 12           |
| EdgeworthBox: Box frame, contract curve, core, endowment, price line and Walrasian equilibrium fully follow the theme                                | 18           | Canvas.add_decomposition: Effect, Marker and Label style objects                                                                             | 16           |
| Theme: Complete theme system: background colour, label scale and defaults for every economic layer                                                   | 29           | EdgeworthBox: Markers, text and axes accept Marker, Label and Axis                                                                           | 18           |
| themes: New paper, monochrome, presentation and dark themes                                                                                          | 29           | <b>v1.7.0</b>                                                                                                                                | (2026-09-25) |
| Config: Configuration files accept every built-in theme and keep the same visual meaning across diagram types                                        | 31           | <b>General:</b> CI tests Python 3.10–3.13, enforces branch coverage, and smoke-tests runnable examples                                       | 2            |
| econ-viz plot: --theme accepts every built-in CLI theme                                                                                              | 32           | Tagged releases publish only after tests pass                                                                                                | 2            |
| <b>v1.10.1</b>                                                                                                                                       | (2026-09-26) | econ-viz: Package management and builds migrated to uv                                                                                       | 3            |
| <b>General:</b> Ships py.typed; CI runs Ruff and Mypy                                                                                                | 2            | All example scripts run from a clean checkout                                                                                                | 4            |
| <b>v1.10.0</b>                                                                                                                                       | (2026-09-26) | econ-viz --version prints the installed package version                                                                                      | 32           |
| econ-viz: Python 3.10 reads TOML configuration files with tomli                                                                                      | 3            | Canvas: Per-figure text and math fonts without changing matplotlib's global configuration                                                    | 7            |
| Figure: Falls back to the active Config when no theme or font is given                                                                               | 14           | Independent axis-label positions, line styles and arrowhead styles for Canvas and Figure                                                     | 12           |
| EdgeworthBox: Falls back to the active Config when no theme or font is given                                                                         | 18           | Stroke: One Stroke controls line width, style, colour and arrowheads across canvases, figures, demand diagrams and Edgeworth boxes           | 12           |
| Config: Config, econ-viz.toml, econ-viz init and plot --config                                                                                       | 31           | CobbDouglas: Fixed utility domains, asymmetric CES expansion paths, Cobb-Douglas limit cases, and satiation optima that leave income unspent | 19           |
| econ-viz init: Configuration-template command and plot --config                                                                                      | 32           | Fixed contour levels when utility is near zero or negative                                                                                   | 19           |
| <b>v1.9.0</b>                                                                                                                                        | (2026-09-26) | comparative_statics: Comparative statics no longer step outside the domain near zero prices or income, or near subsistence constraints       | 26           |
| Canvas: Axes, origin, titles and other text elements accept a Label                                                                                  | 7            | <b>v1.6.0</b>                                                                                                                                | (2026-04-24) |
| Stroke: New opacity field                                                                                                                            | 12           |                                                                                                                                              |              |

|                                                                                                                                                                                                                                                         |    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>General:</b> Generated files under <code>examples/output/</code> are no longer tracked .....                                                                                                                                                         | 2  |
| <b>Canvas:</b> Default indifference-curve line width reduced from 2.0 to 1.8 .....                                                                                                                                                                      | 12 |
| <b>themes.default:</b> Colour-blind-friendly default palette, with <code>COLORBLIND_CYCLE_RGB</code> and <code>COLORBLIND_CYCLE_HEX</code> .....                                                                                                        | 29 |
| <b>Canvas.save:</b> TikZ axis regression tests no longer depend on colour indices .....                                                                                                                                                                 | 32 |
| <b>Animator:</b> Animation examples moved to <code>examples/scripts/animation.py</code> .....                                                                                                                                                           | 36 |
| <b>v1.5.0</b> (2026-04-21)                                                                                                                                                                                                                              |    |
| <b>General:</b> New example scripts for decomposition, demand diagrams, PCC/ICC paths, multi-panel layouts and TikZ export .....                                                                                                                        | 2  |
| Regression tests for TikZ export and price-effect decomposition .....                                                                                                                                                                                   | 2  |
| <b>econ-viz:</b> Notebook install flow on Colab is restart-safe .....                                                                                                                                                                                   | 4  |
| <b>Figure:</b> Better placement and visibility of axis labels in shared-panel layouts .....                                                                                                                                                             | 14 |
| <b>decompose_price_effect:</b> Price-effect decomposition: <code>decompose_price_effect()</code> , <code>PriceEffectDecomposition</code> and <code>Canvas.add_decomposition()</code> , drawing bundles A/B/C with substitution and income effects ..... | 16 |
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| <b>v1.4.0</b> (2026-04-09)                                                                                                                                                                                                                              |    |
| <b>General:</b> Tests for <code>Animator</code> and <code>WidgetViewer</code> .....                                                                                                                                                                     | 2  |
| <b>econ-viz[extras]:</b> Optional extras animation, interactive and all .....                                                                                                                                                                           | 3  |
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| <b>v1.3.1</b> (2026-04-03)                                                                                                                                                                                                                              |    |
| <b>General:</b> Package-root exports load lazily, with the public API unchanged .....                                                                                                                                                                   | 2  |
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